

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line

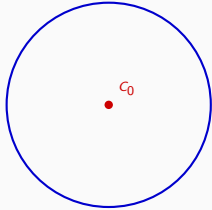
$C_0 \rightarrow$ Initial Center Point

$AA' \rightarrow$ Directing Line

Axis $\rightarrow$ Center Line



Step 1: Draw Generating Circle & Directing Line

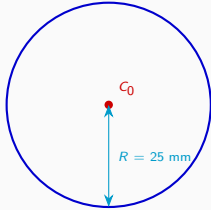


$C_0 \rightarrow$ Initial Center Point

$AA' \rightarrow$ Directing Line

Axis $\rightarrow$ Center Line

Step 1: Draw Generating Circle & Directing Line



C_0 → Initial Center Point

AA' → Directing Line

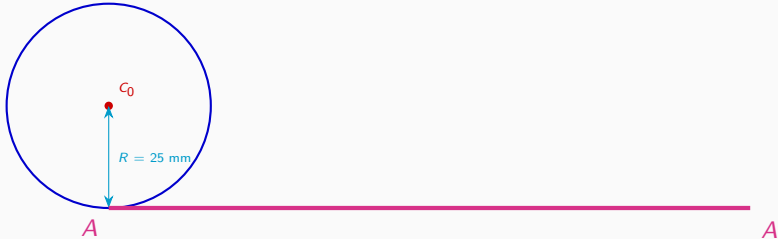
Axis → Center Line

Step 1: Draw Generating Circle & Directing Line

C_0 → Initial Center Point

AA' → Directing Line

Axis → Center Line



Step 1: Draw Generating Circle & Directing Line

C_0 → Initial Center Point

AA' → Directing Line

Axis → Center Line



Step 1: Draw Generating Circle & Directing Line

C_0 → Initial Center Point

AA' → Directing Line

Axis → Center Line



Step 1: Draw Generating Circle & Directing Line

C_0 → Initial Center Point

AA' → Directing Line

Axis → Center Line

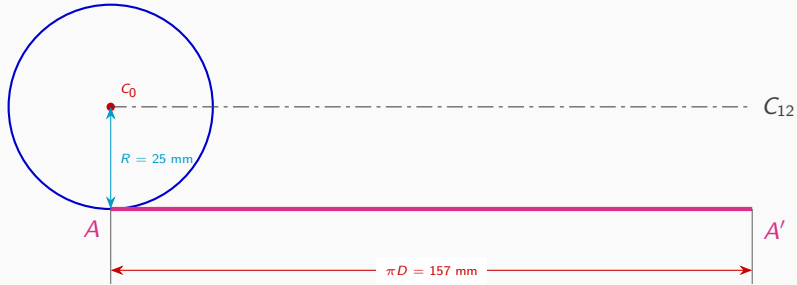


Step 1: Draw Generating Circle & Directing Line

C_0 → Initial Center Point

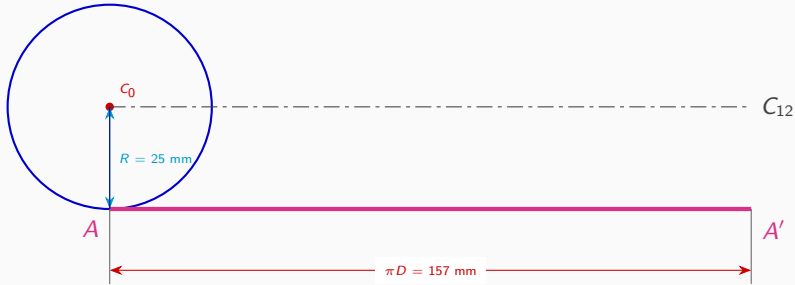
AA' → Directing Line

Axis → Center Line



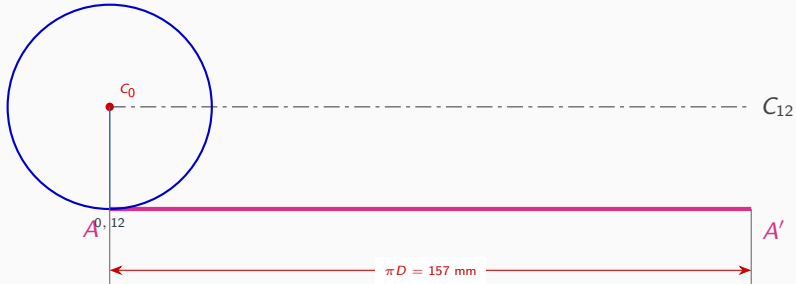
Step 1: Draw Generating Circle & Directing Line

C_0 → Initial Center Point
 AA' → Directing Line
Axis → Center Line



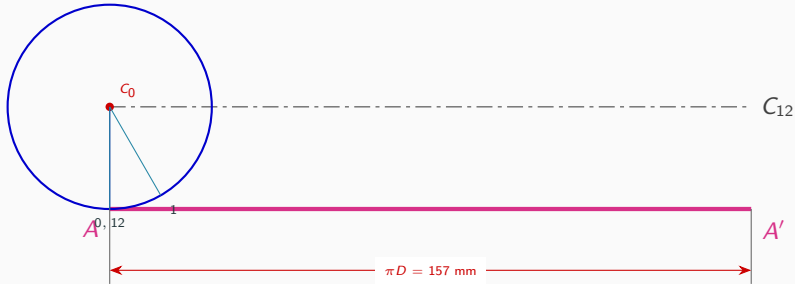
Step 2: Divide Generating Circle into 12 Equal Parts

AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions



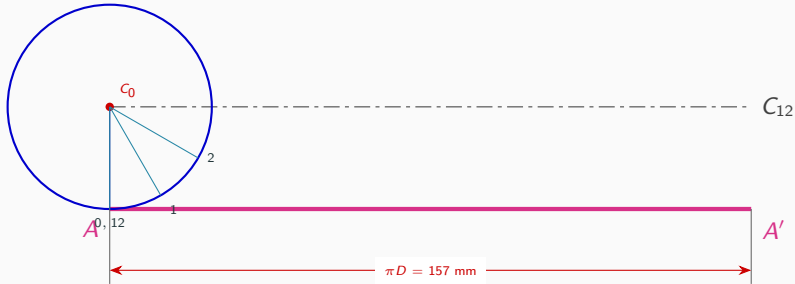
Step 2: Divide Generating Circle into 12 Equal Parts

AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions



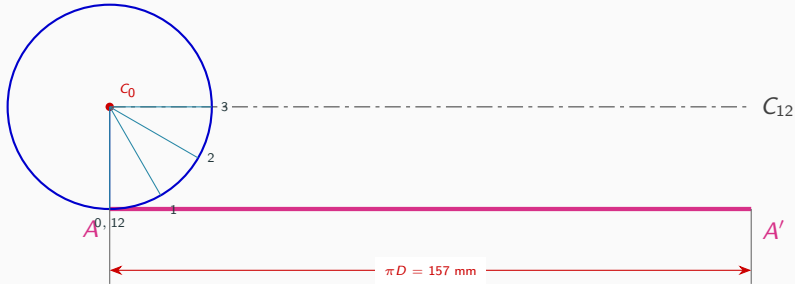
Step 2: Divide Generating Circle into 12 Equal Parts

AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions



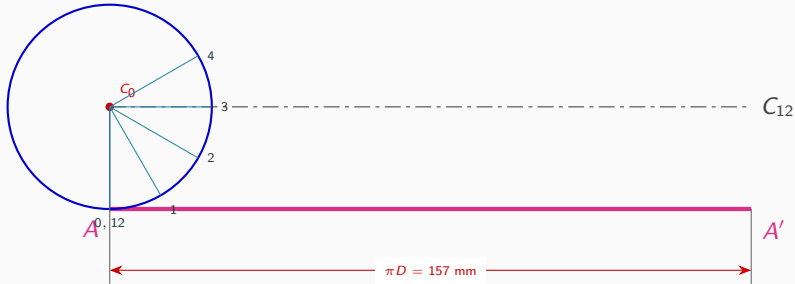
Step 2: Divide Generating Circle into 12 Equal Parts

AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions



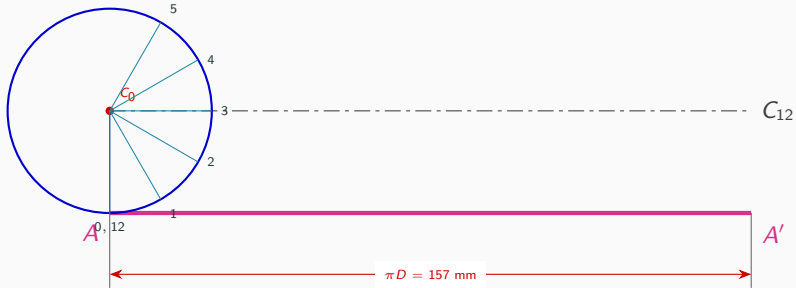
Step 2: Divide Generating Circle into 12 Equal Parts

AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions

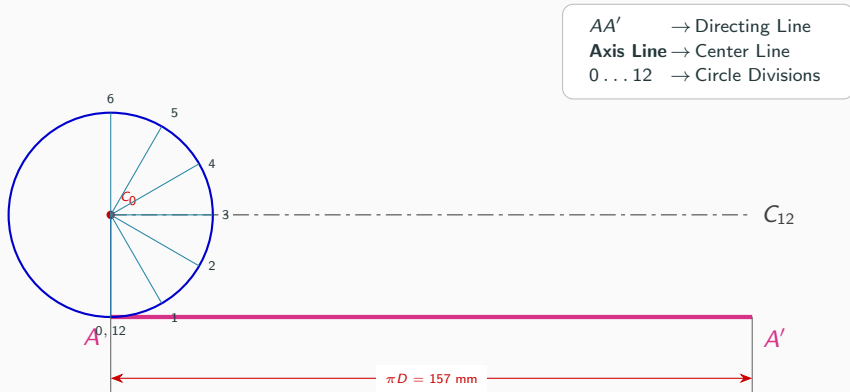


Step 2: Divide Generating Circle into 12 Equal Parts

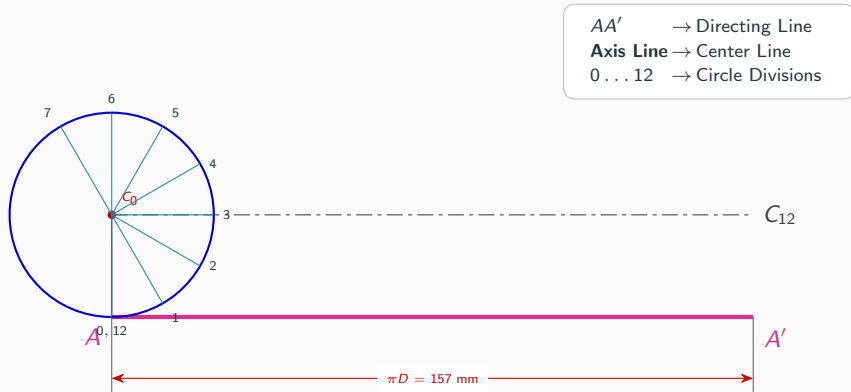
AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions



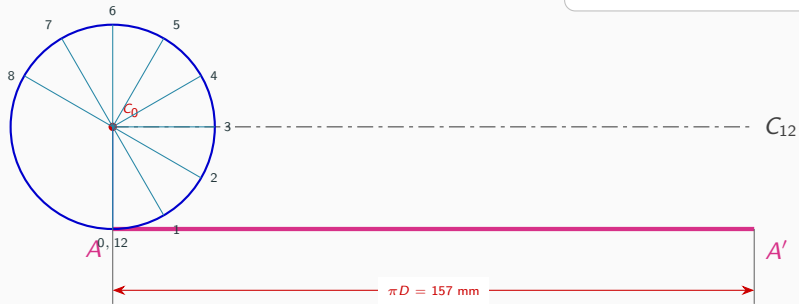
Step 2: Divide Generating Circle into 12 Equal Parts



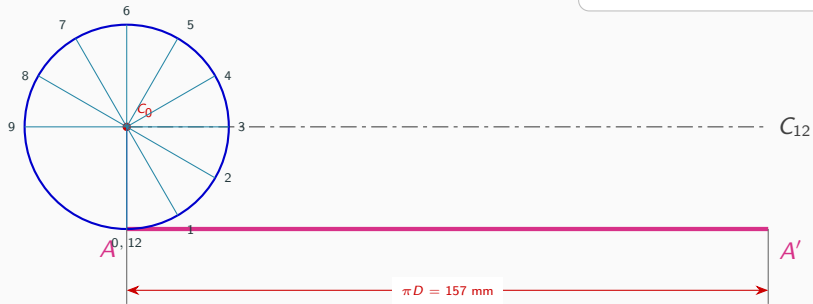
Step 2: Divide Generating Circle into 12 Equal Parts



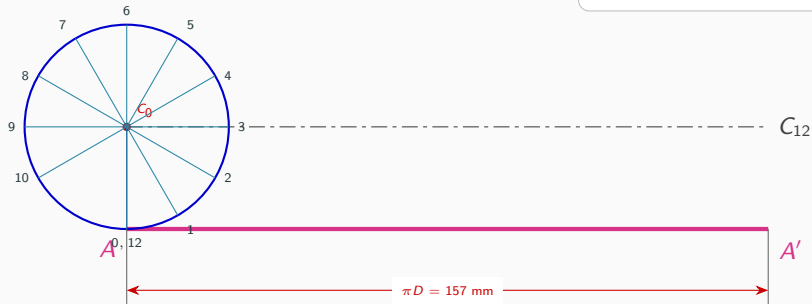
Step 2: Divide Generating Circle into 12 Equal Parts



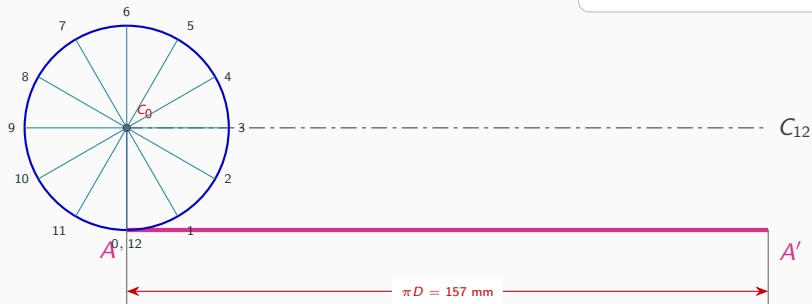
Step 2: Divide Generating Circle into 12 Equal Parts



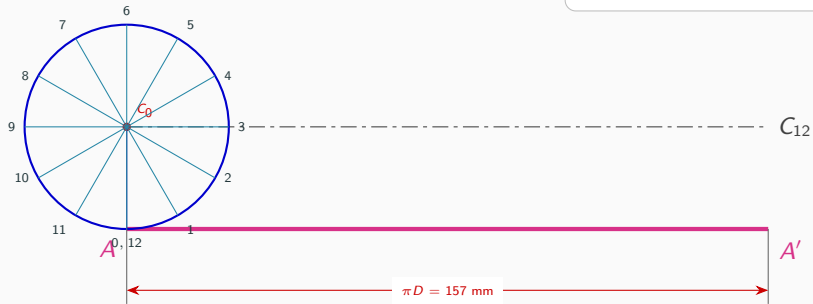
Step 2: Divide Generating Circle into 12 Equal Parts



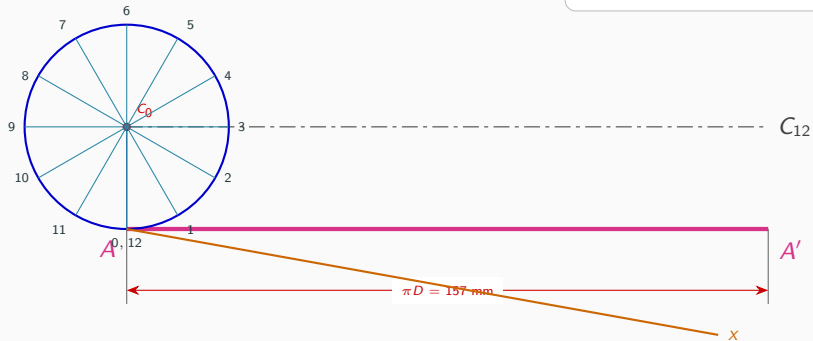
Step 2: Divide Generating Circle into 12 Equal Parts



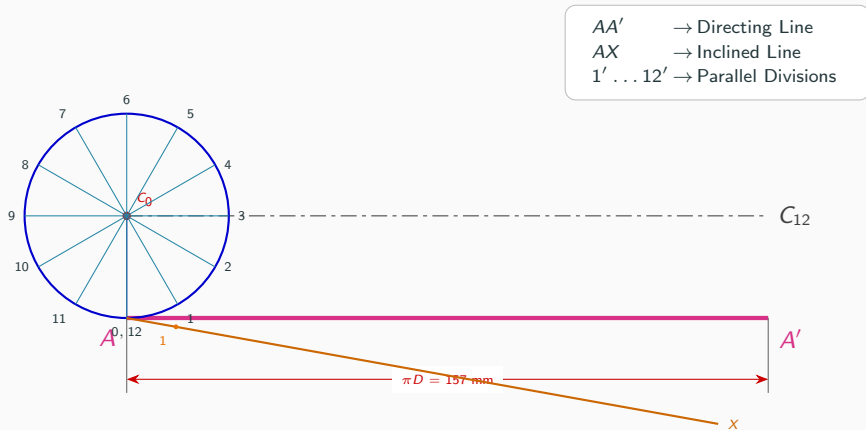
Step 3: Divide Directing Line using Line Division Method



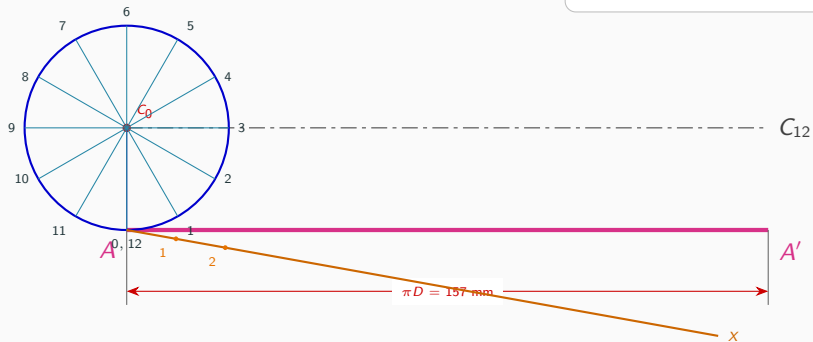
Step 3: Divide Directing Line using Line Division Method



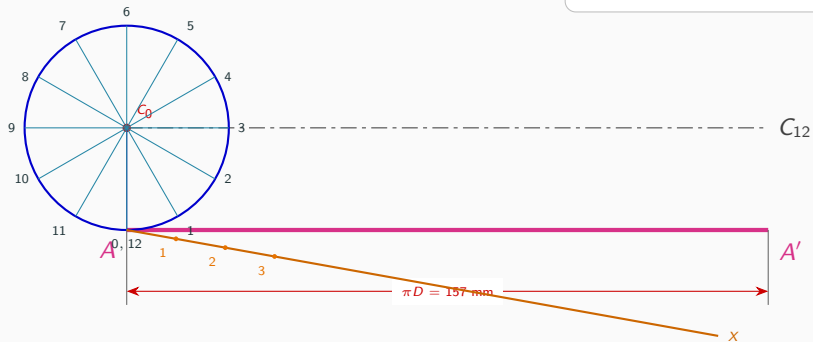
Step 3: Divide Directing Line using Line Division Method



Step 3: Divide Directing Line using Line Division Method

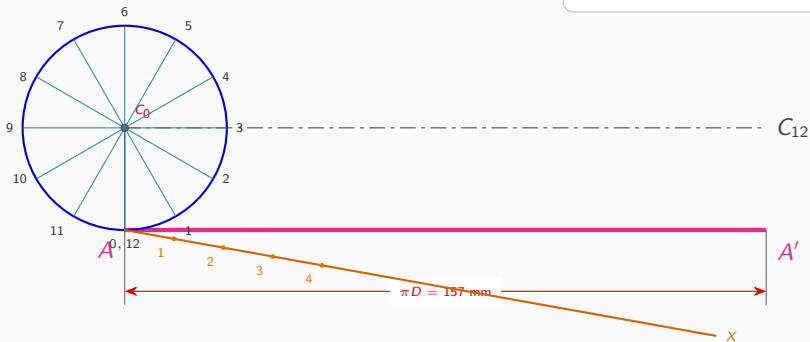


Step 3: Divide Directing Line using Line Division Method



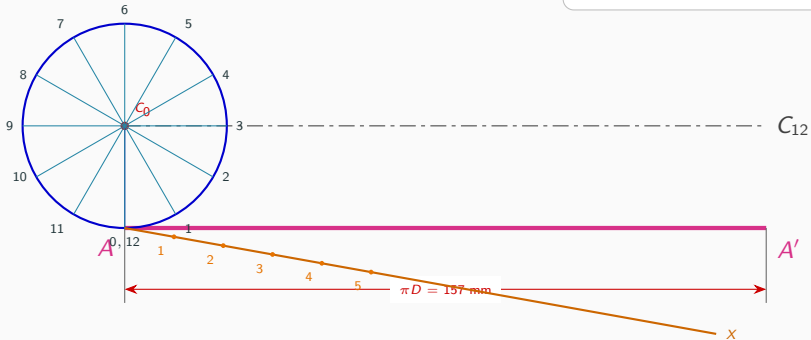
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions

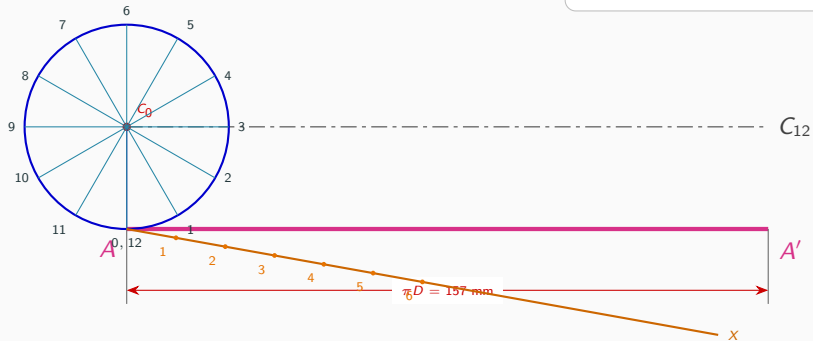


Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions

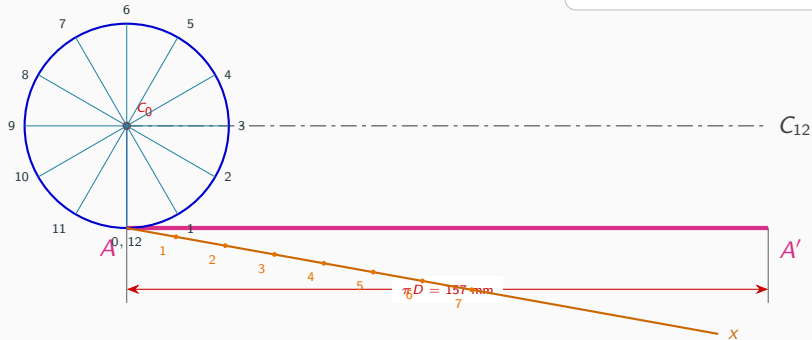


Step 3: Divide Directing Line using Line Division Method



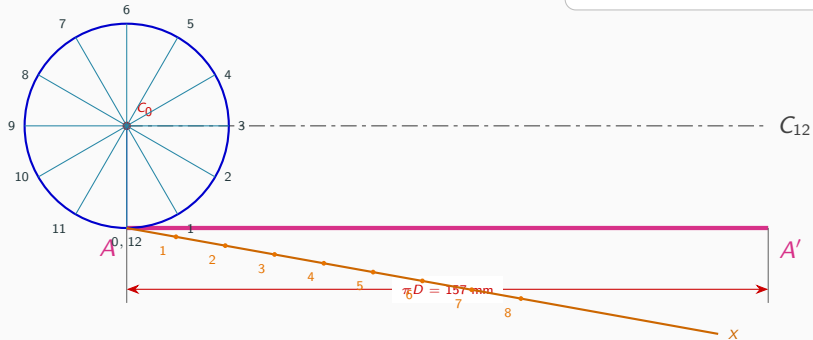
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions

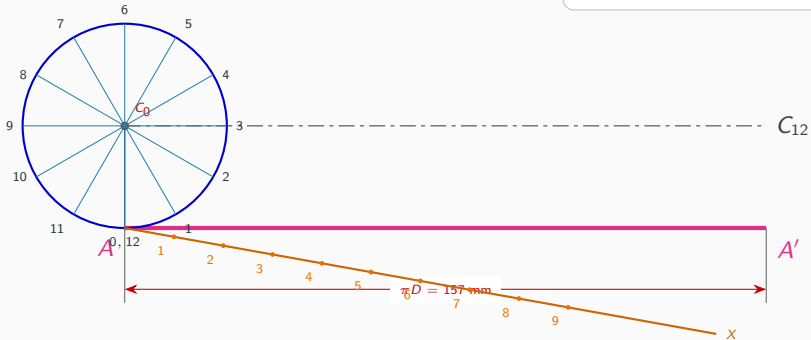


Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions

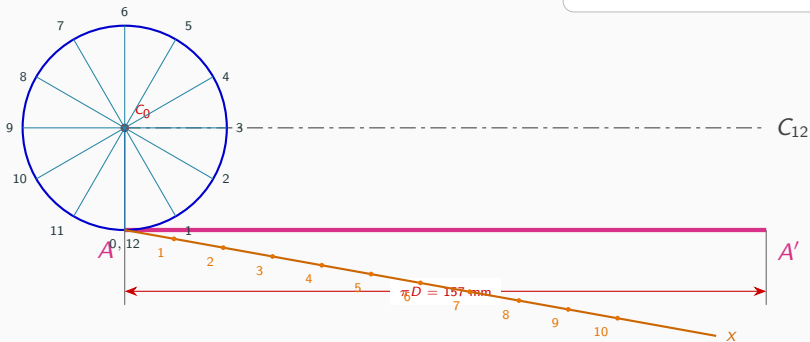


Step 3: Divide Directing Line using Line Division Method



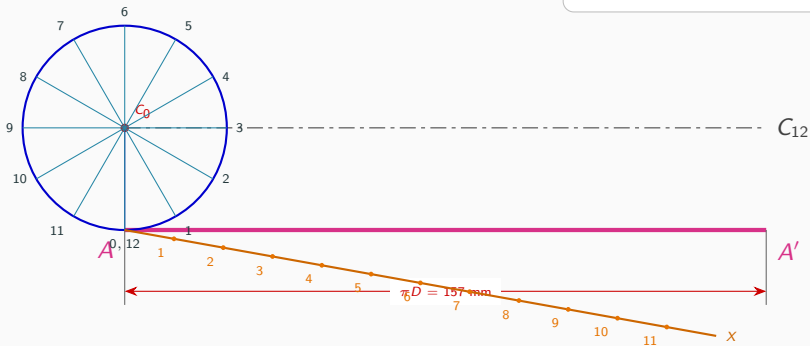
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



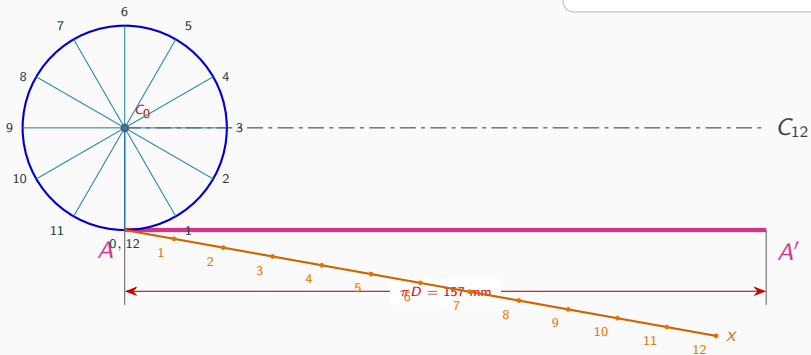
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



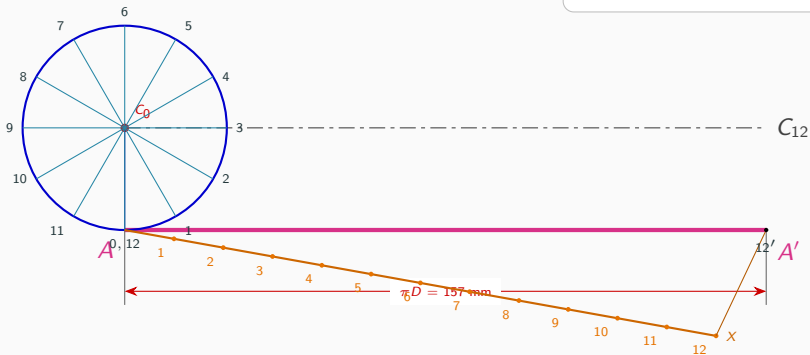
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



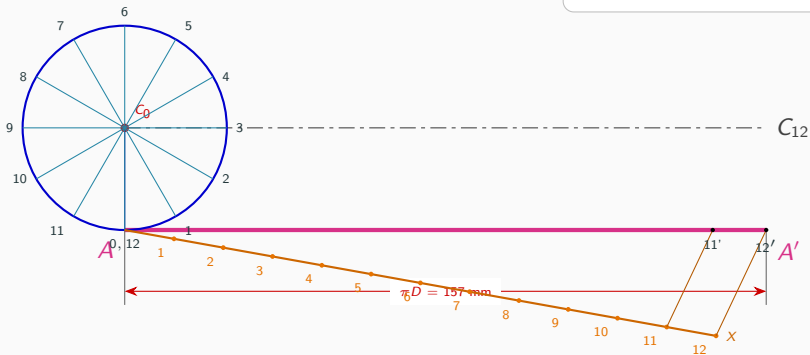
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



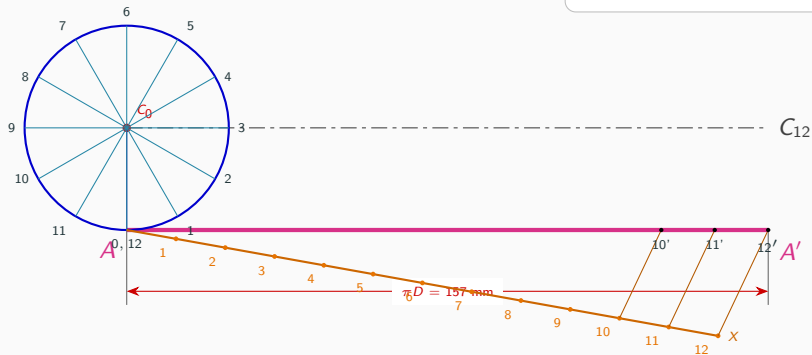
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



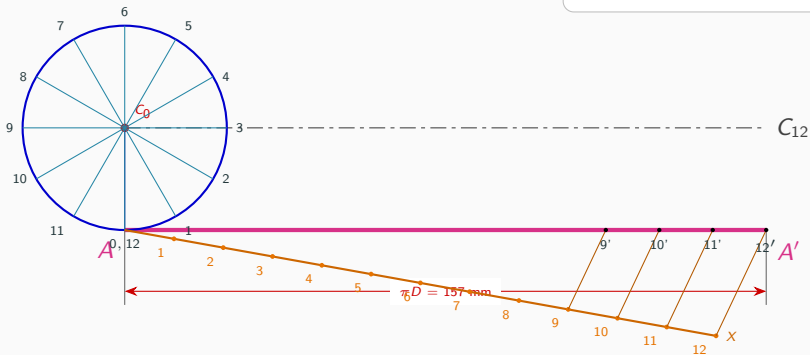
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



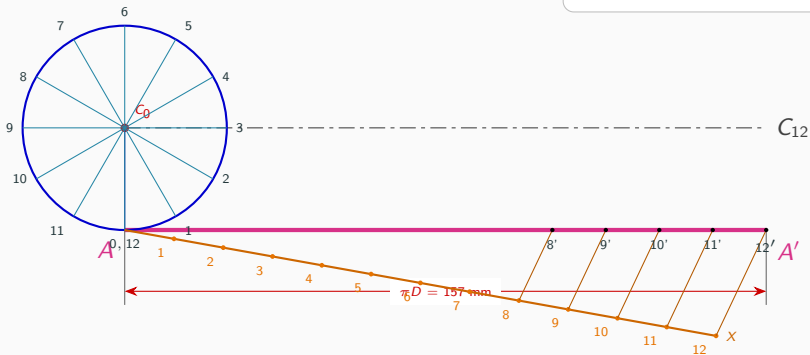
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



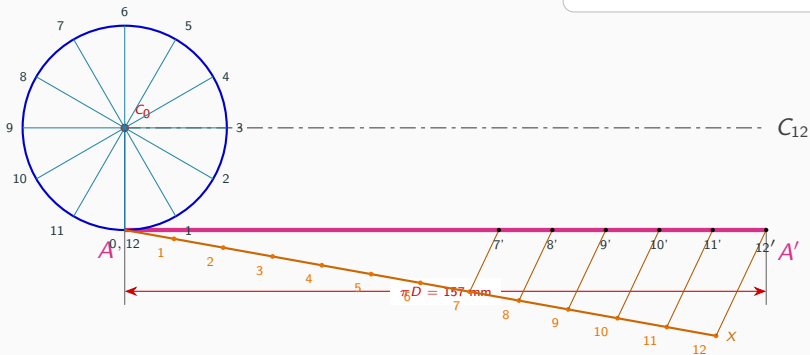
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



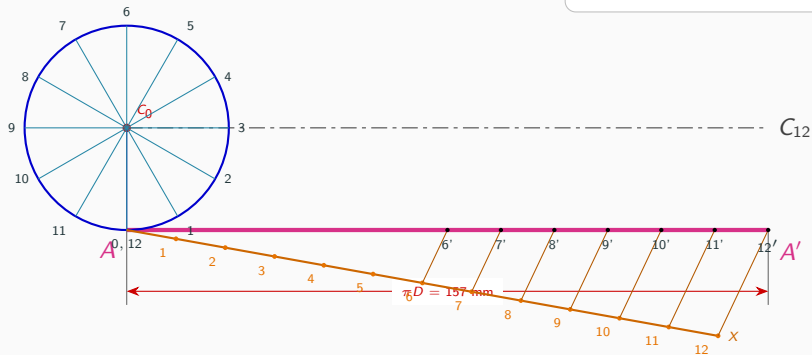
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



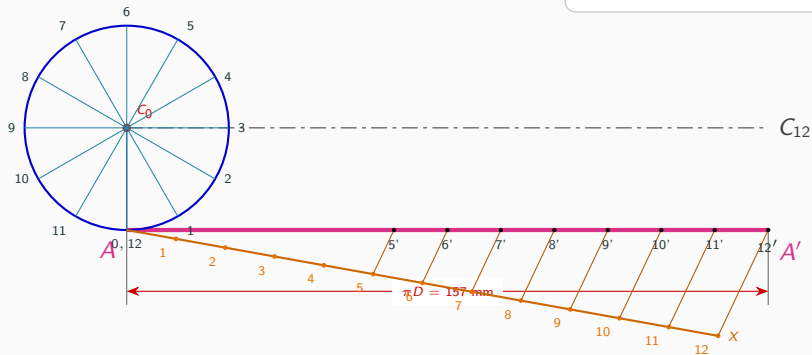
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



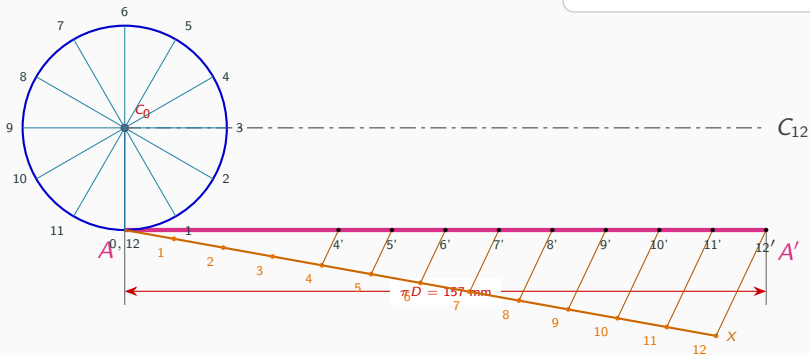
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



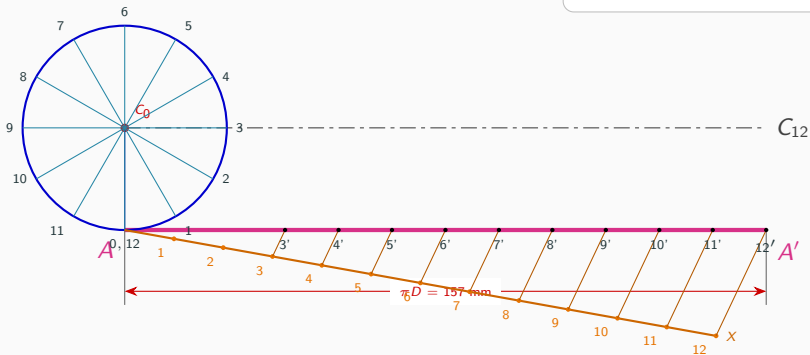
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



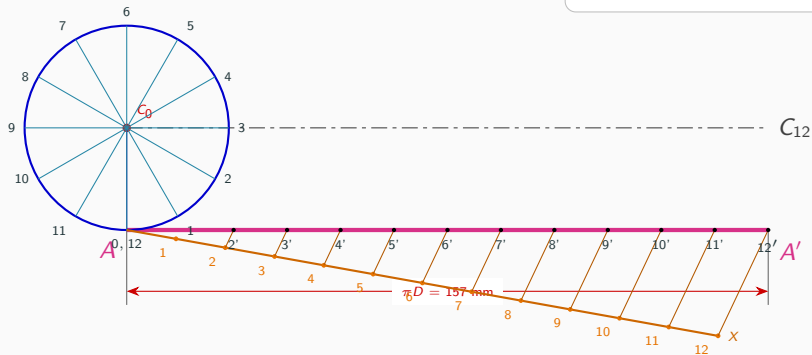
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



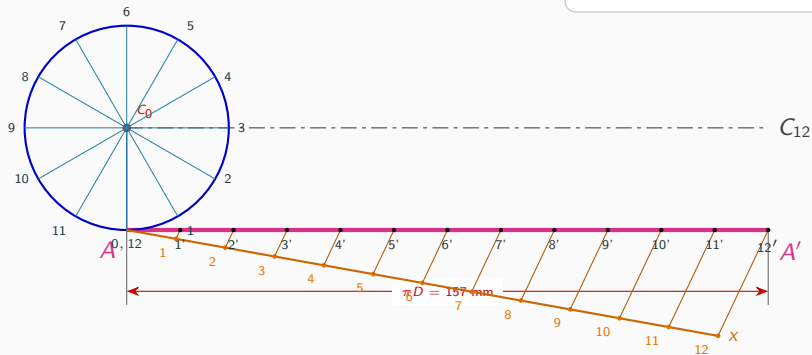
Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions

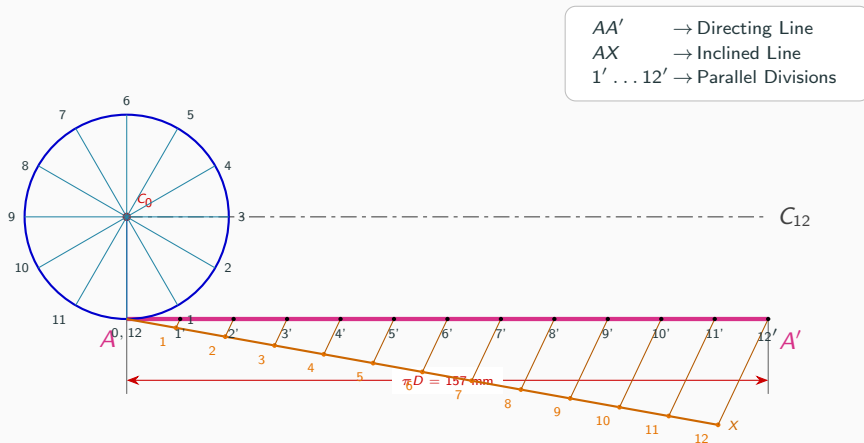


Step 3: Divide Directing Line using Line Division Method

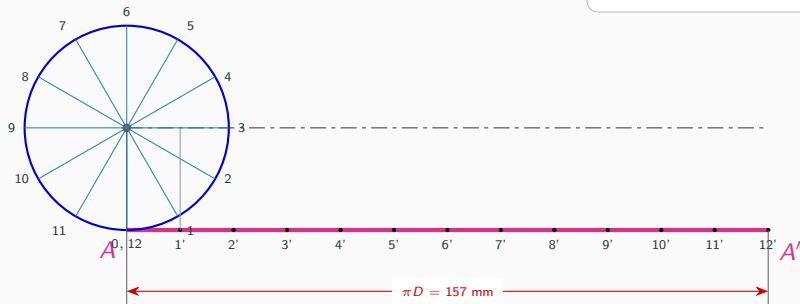
AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



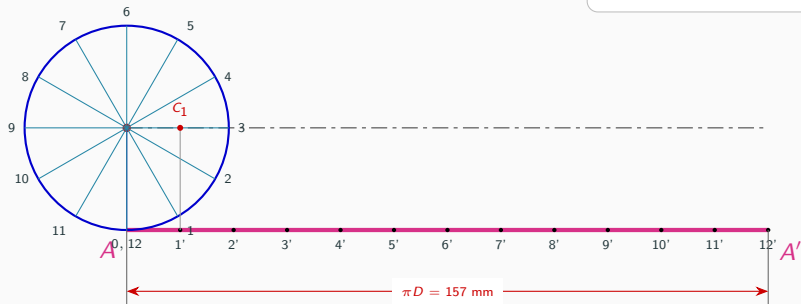
Step 3: Divide Directing Line using Line Division Method



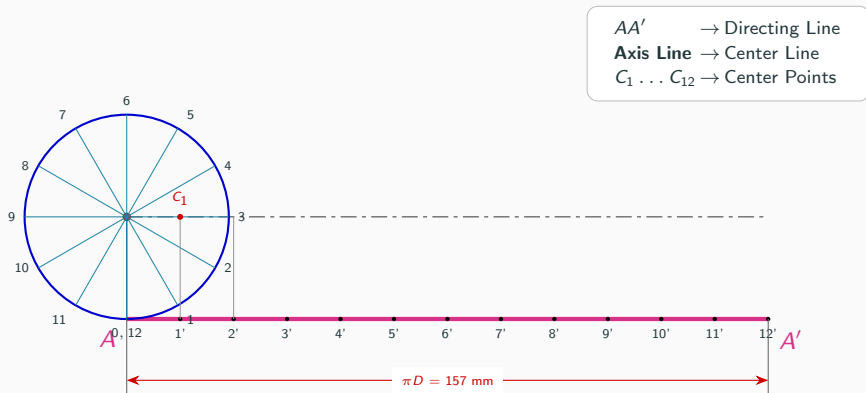
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



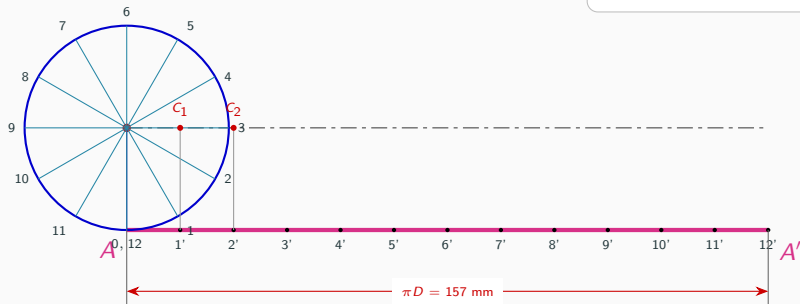
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



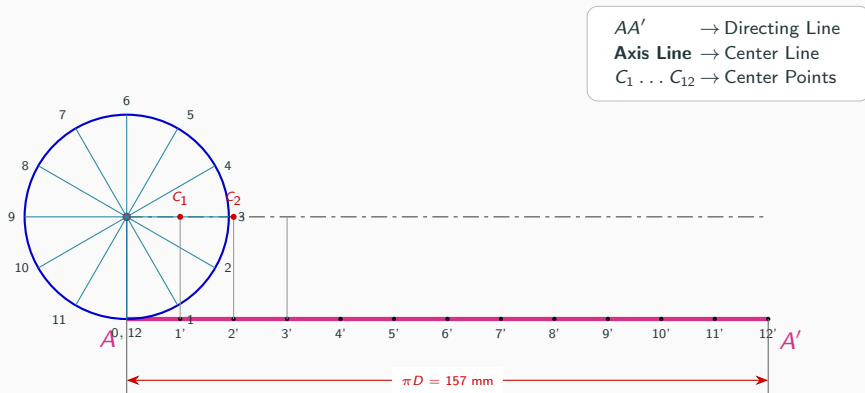
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



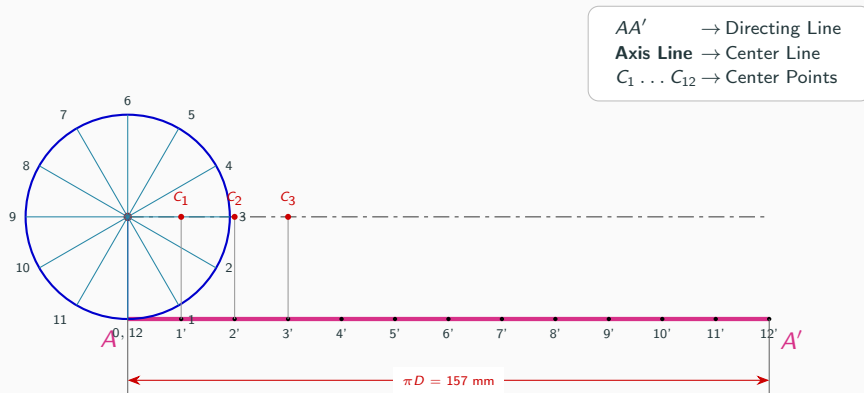
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



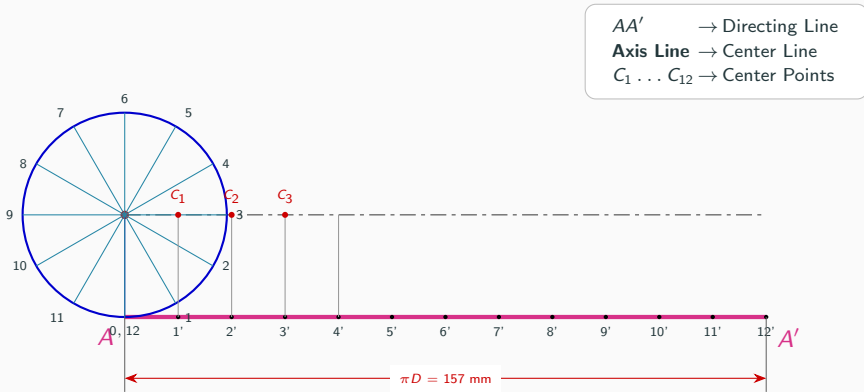
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



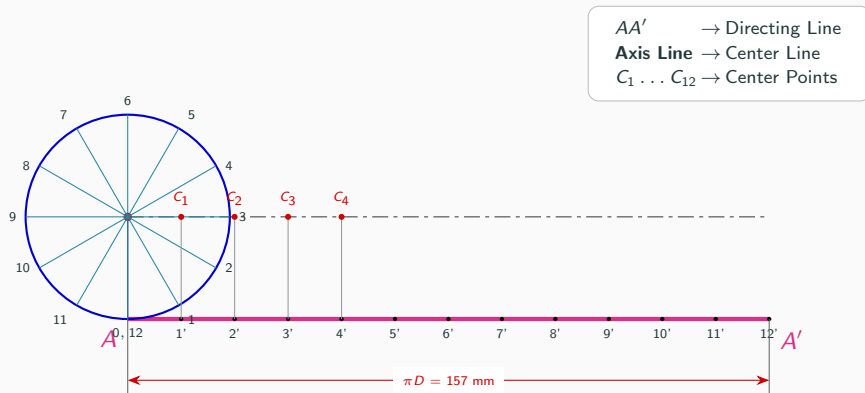
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



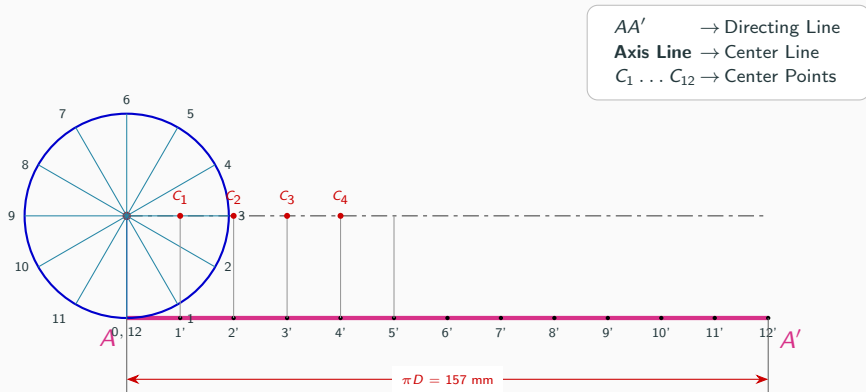
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



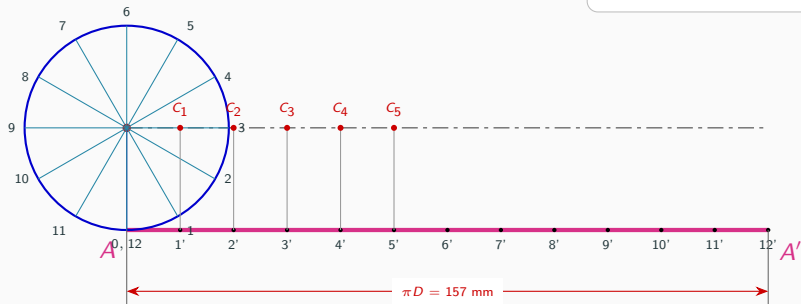
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



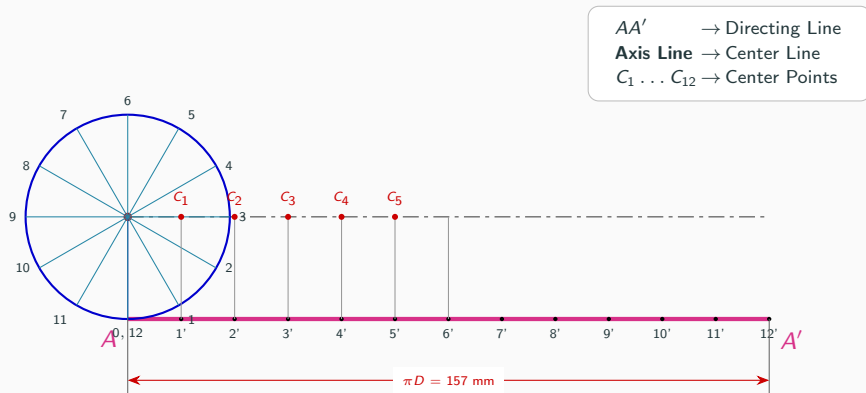
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



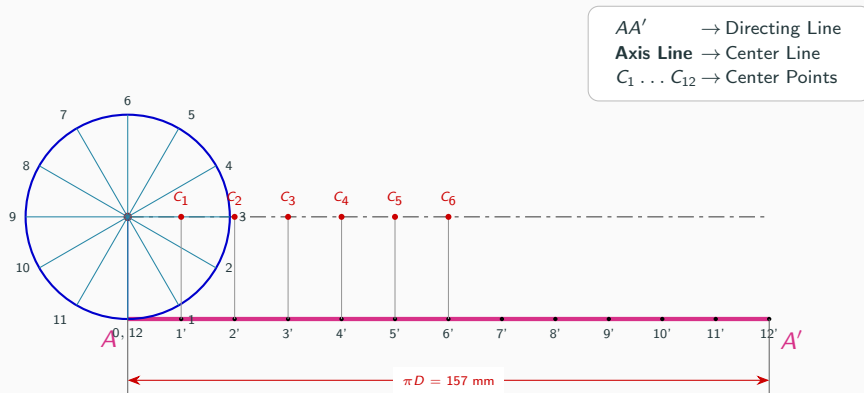
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



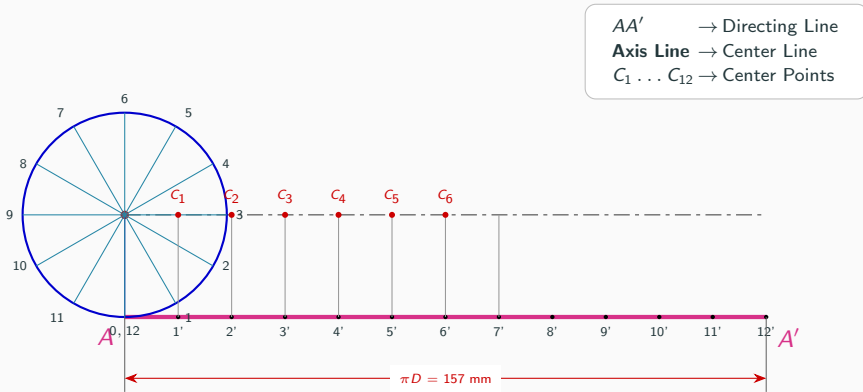
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



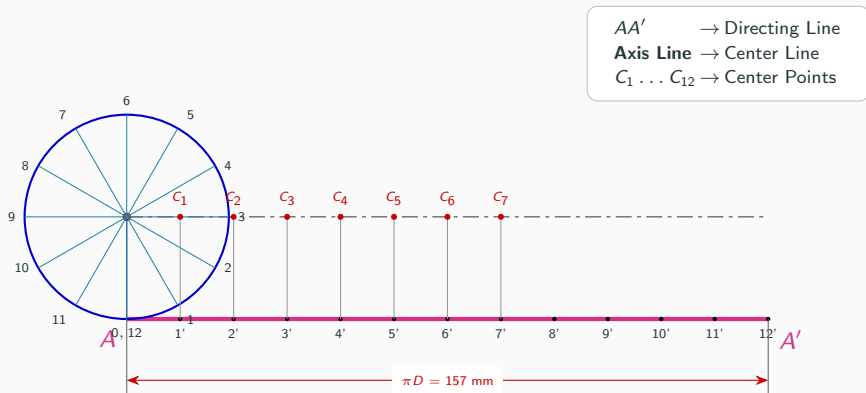
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



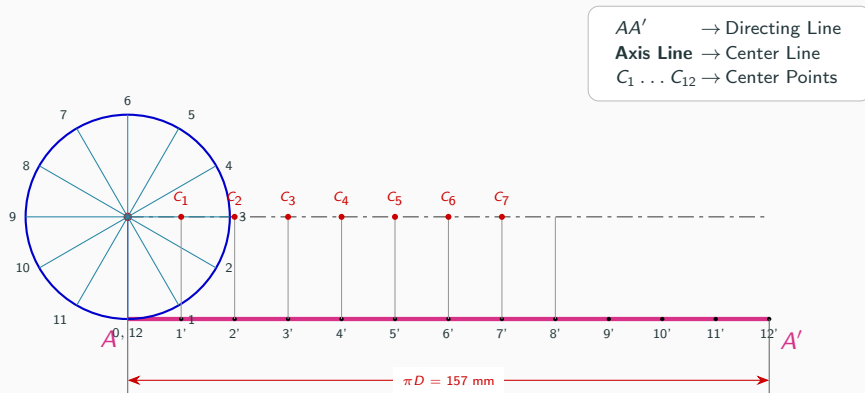
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



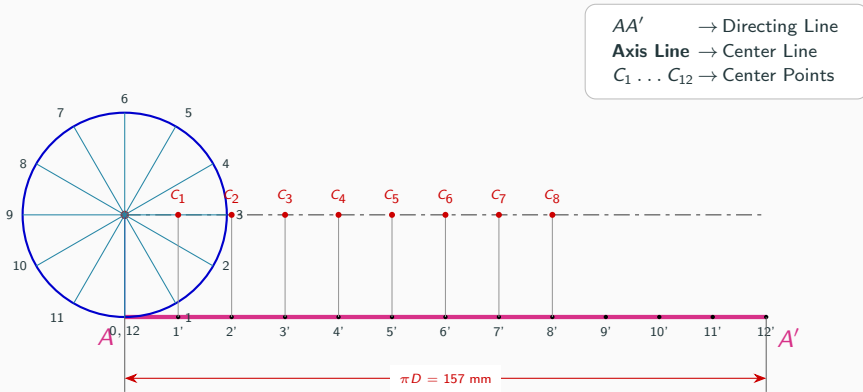
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



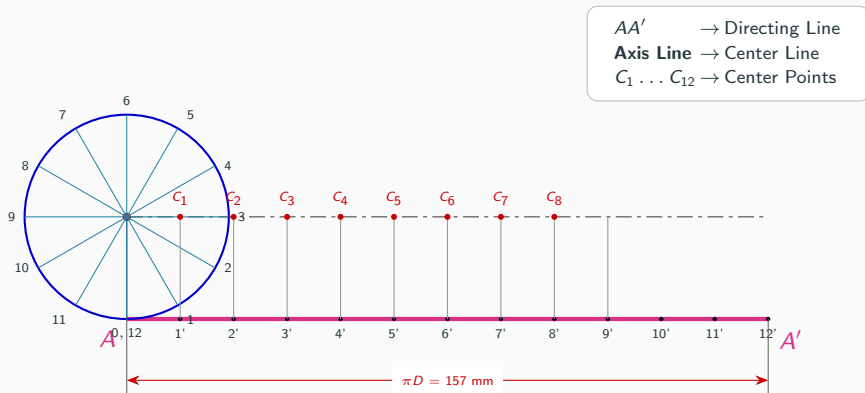
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



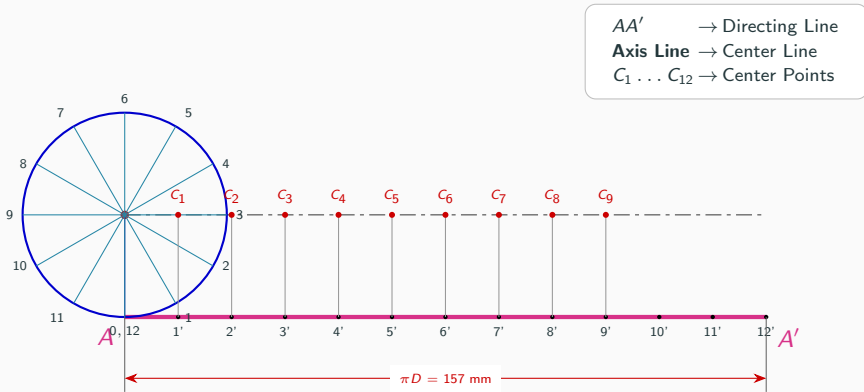
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



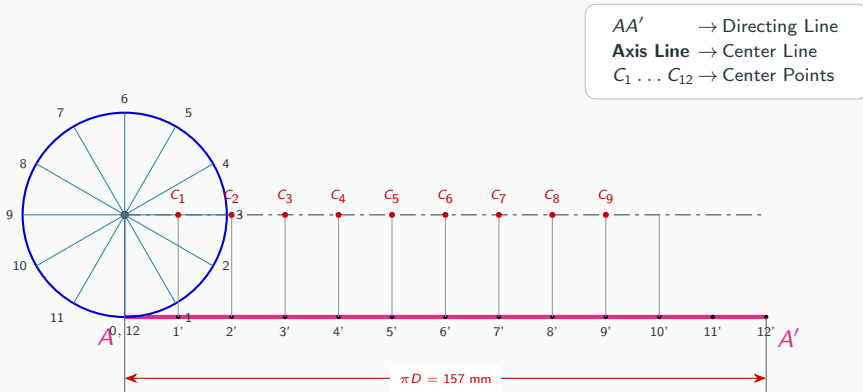
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



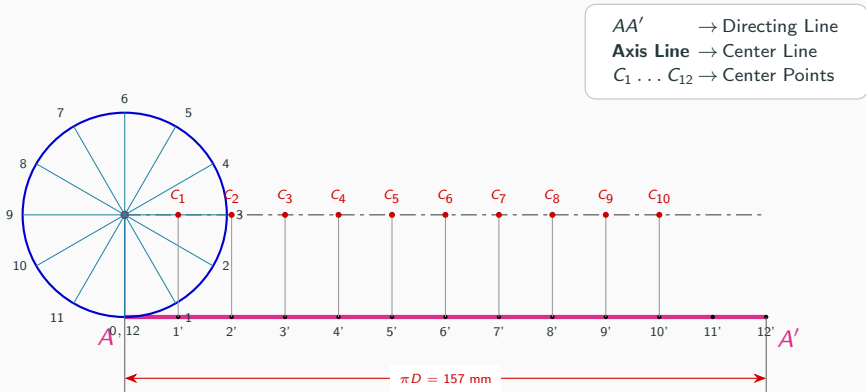
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



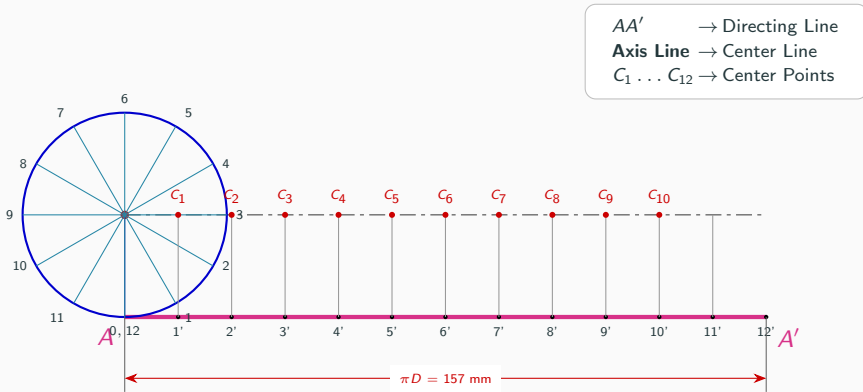
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



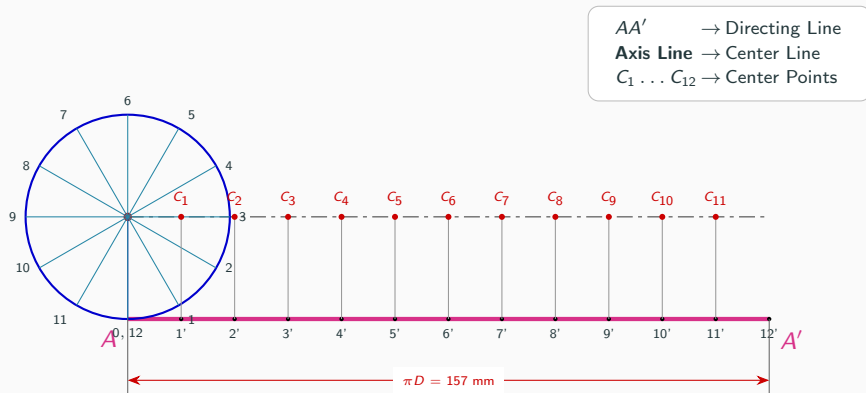
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



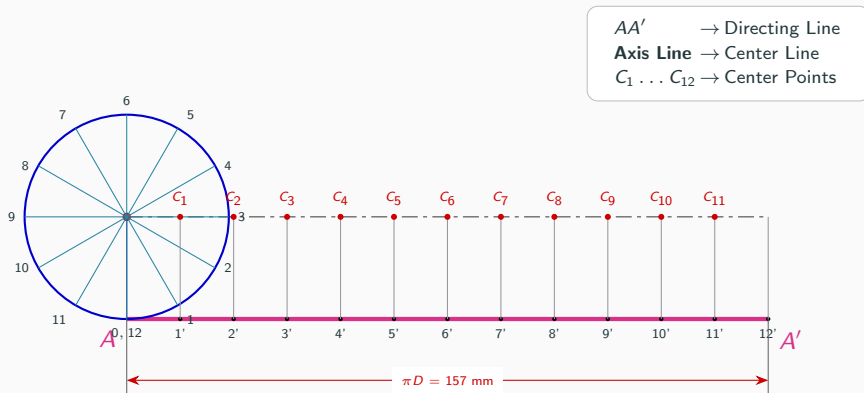
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



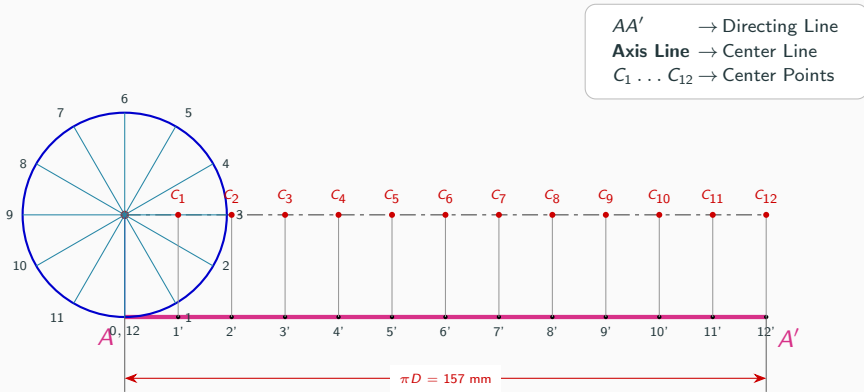
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



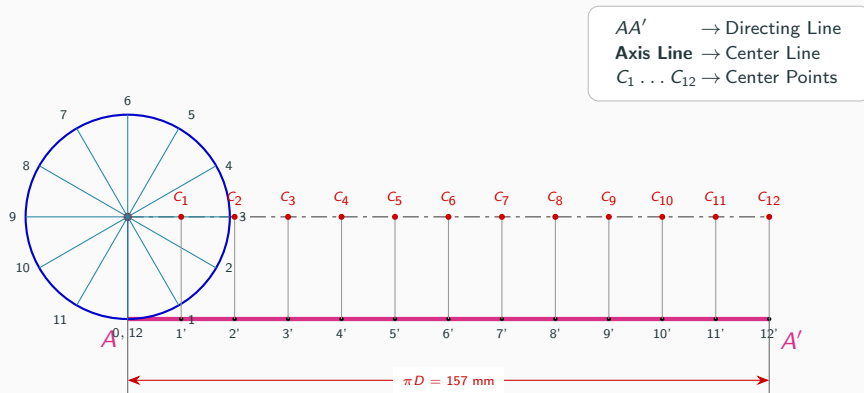
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$

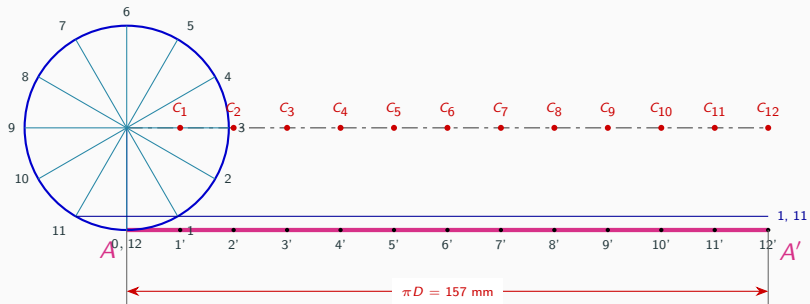


Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



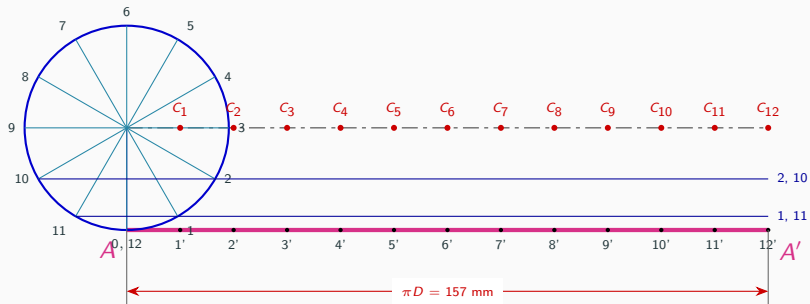
Step 5: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines
 $C_1 \dots C_{12}$ → Retained Centers



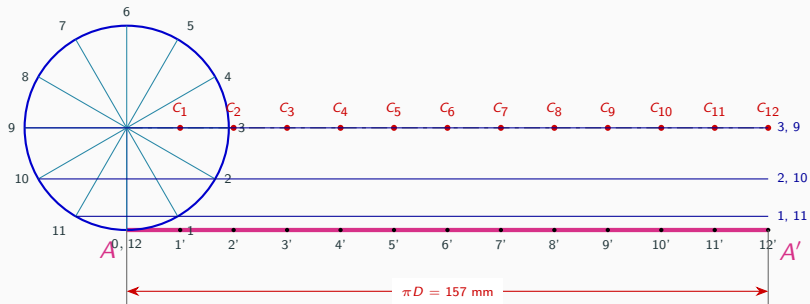
Step 5: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines
 $C_1 \dots C_{12}$ → Retained Centers



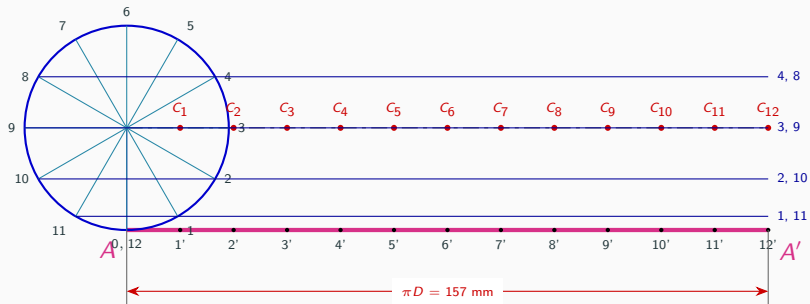
Step 5: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines
 $C_1 \dots C_{12}$ → Retained Centers



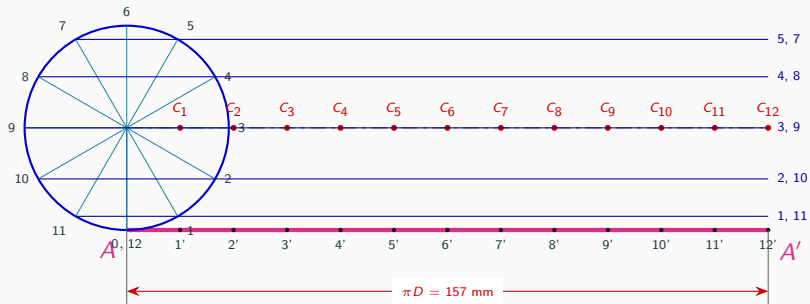
Step 5: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines
 $C_1 \dots C_{12}$ → Retained Centers

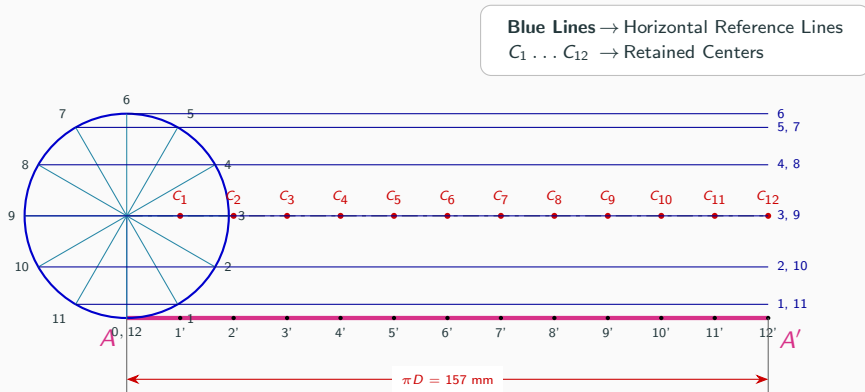


Step 5: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines
 $C_1 \dots C_{12}$ → Retained Centers

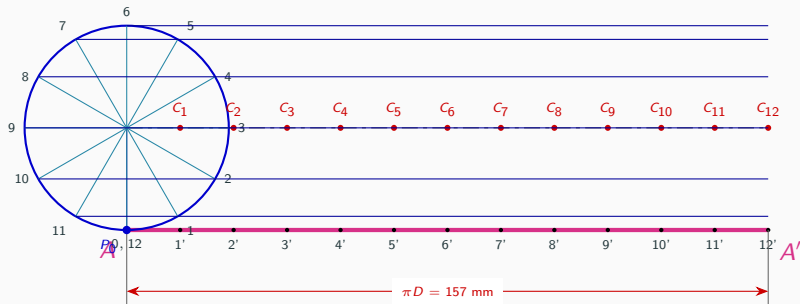


Step 5: Draw Horizontal Reference Lines



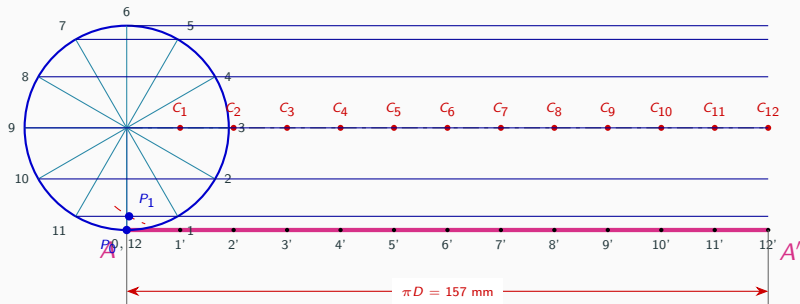
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points



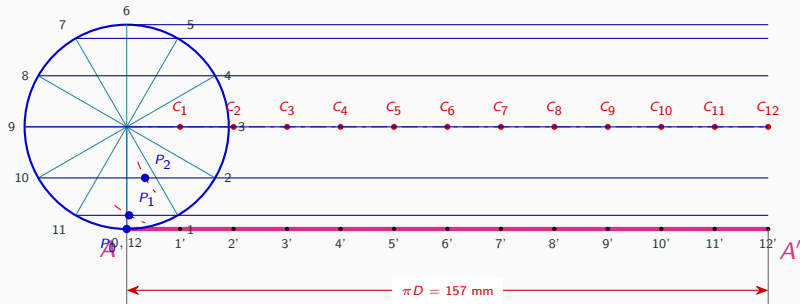
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points

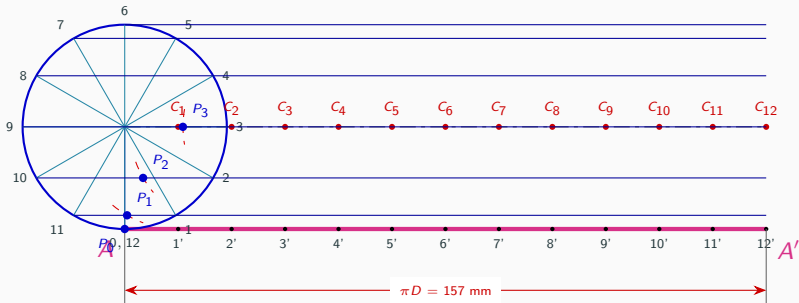


Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points

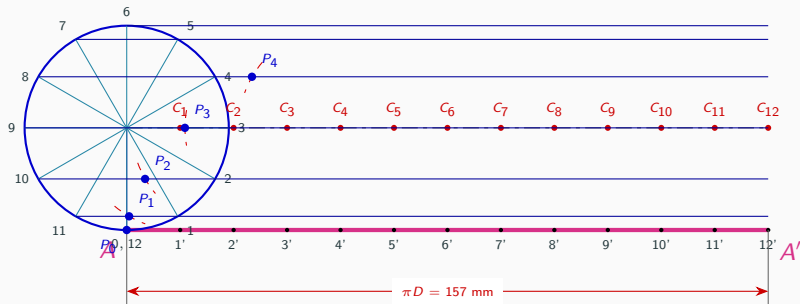


Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$$\begin{aligned} C_1 \dots C_{12} &\rightarrow \text{Retained Centers} \\ P_1 \dots P_{12} &\rightarrow \text{Cutting Arcs/Points} \end{aligned}$$


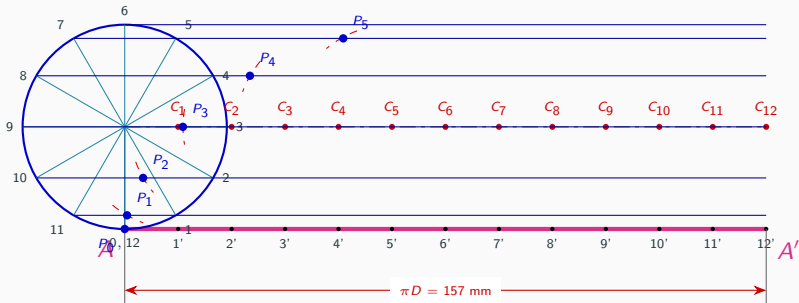
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points



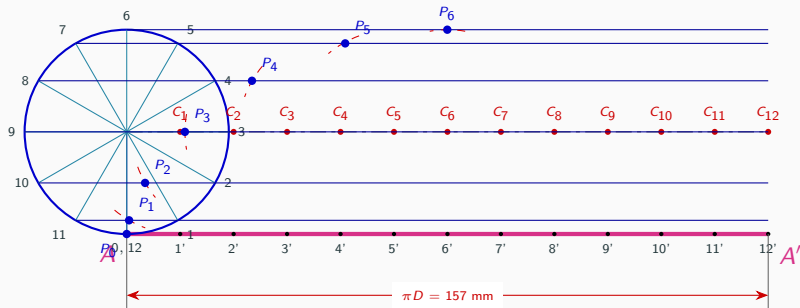
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points



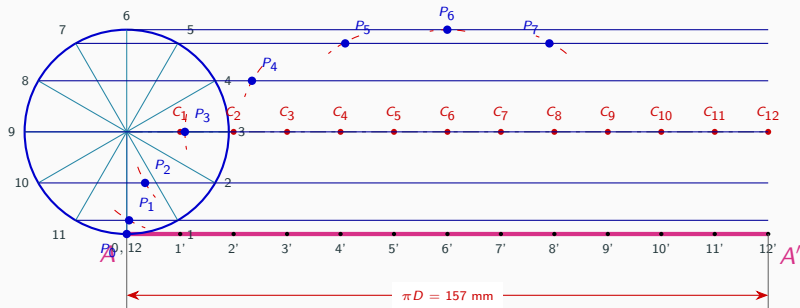
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points

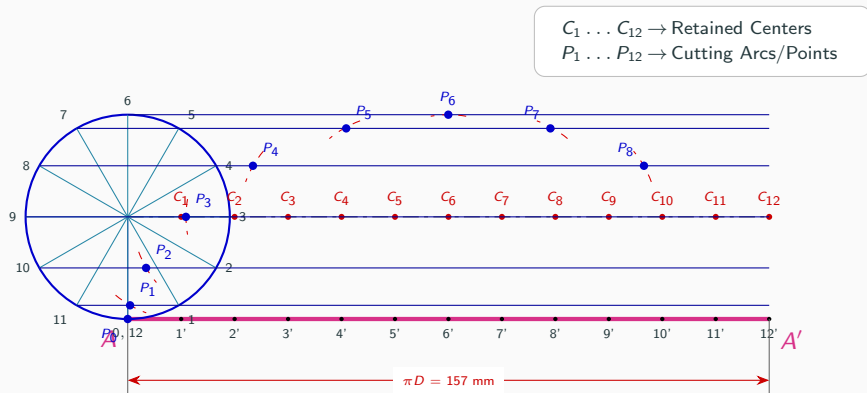


Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

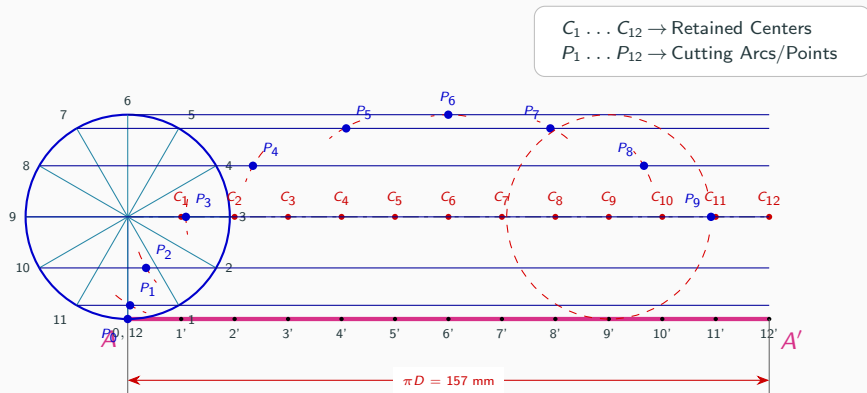
$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points



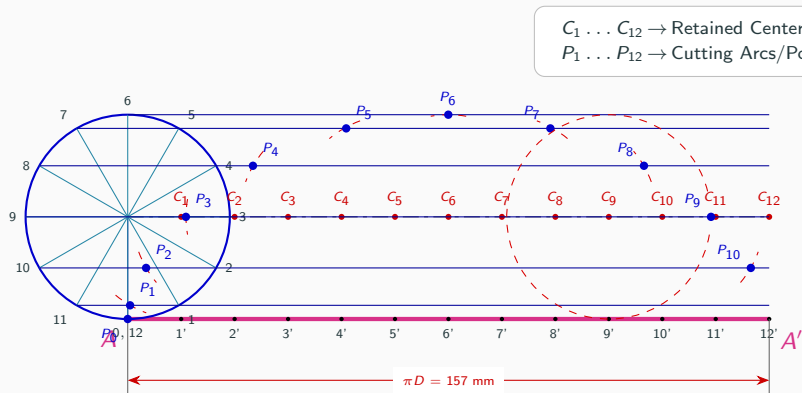
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



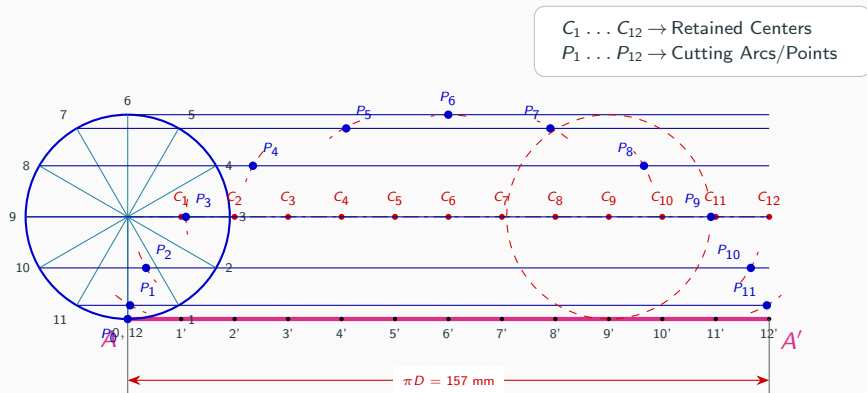
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



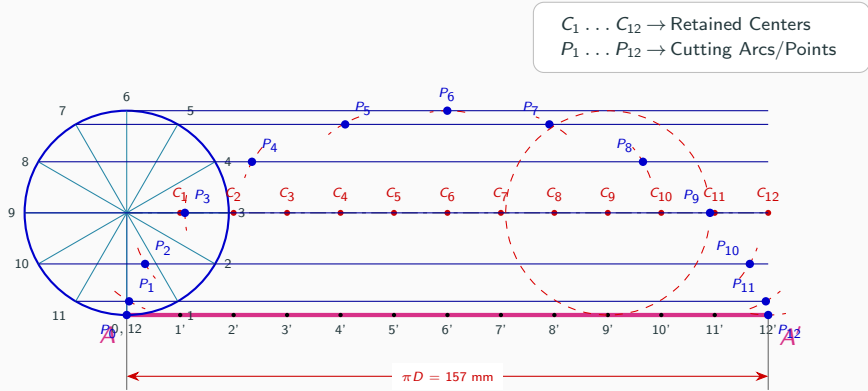
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



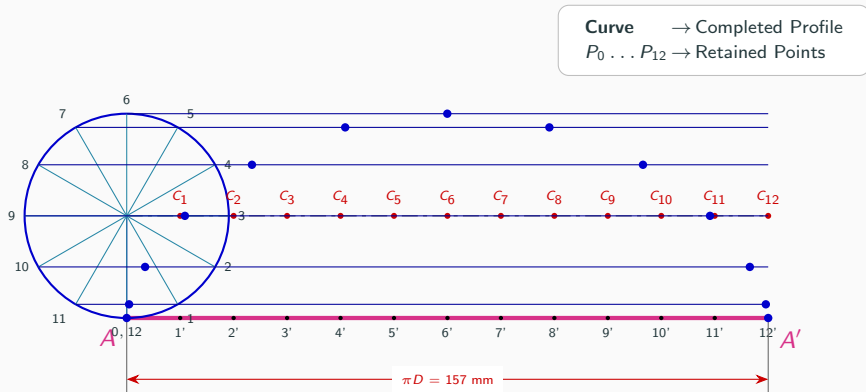
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



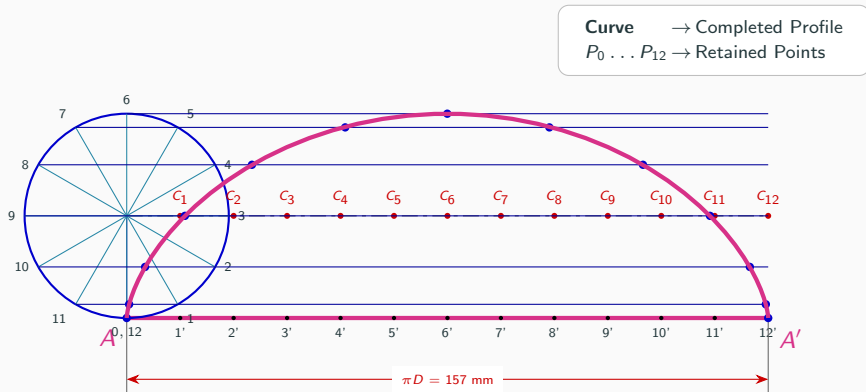
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



Step 7: Smoothly Join Points to Form Cycloid Curve



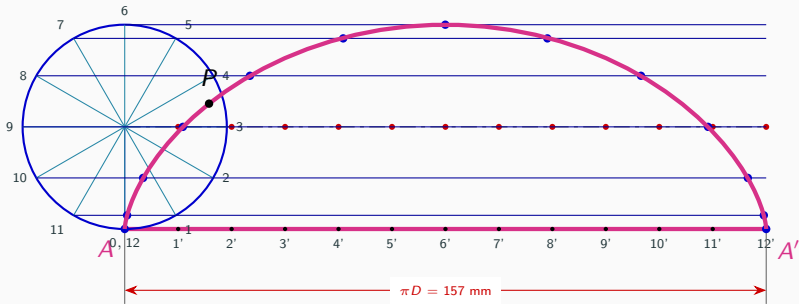
Step 7: Smoothly Join Points to Form Cycloid Curve



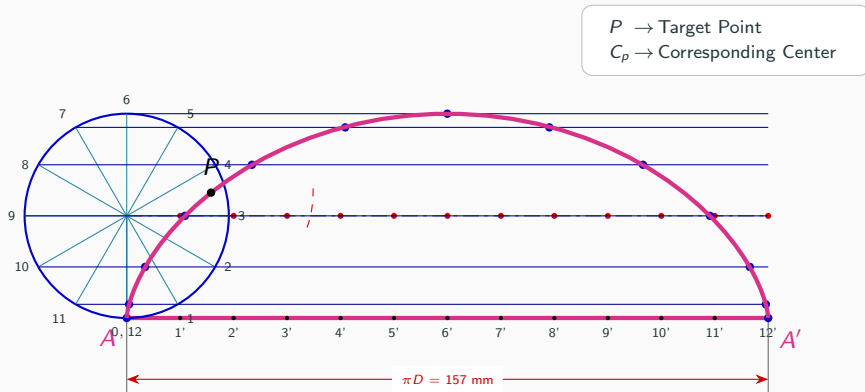
Step 8: Mark Point P & Locate Center C_p

$$P \rightarrow \text{Target Point}$$

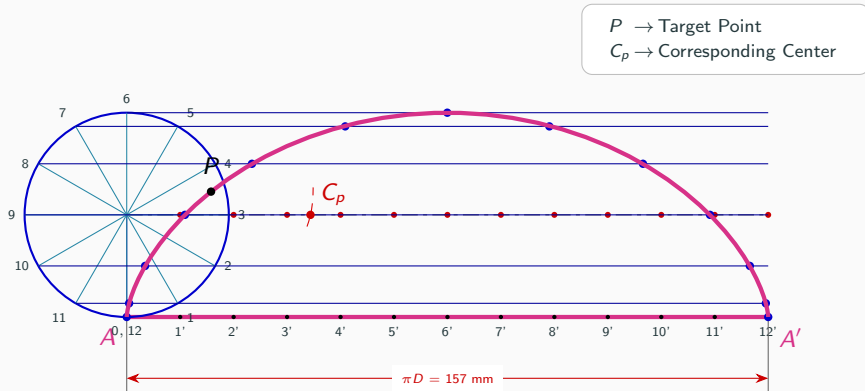
$C_p \rightarrow$ Corresponding Center



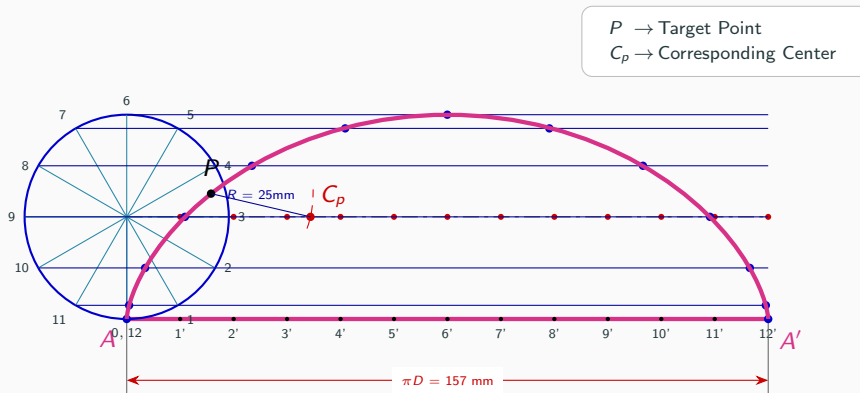
Step 8: Mark Point P & Locate Center C_p



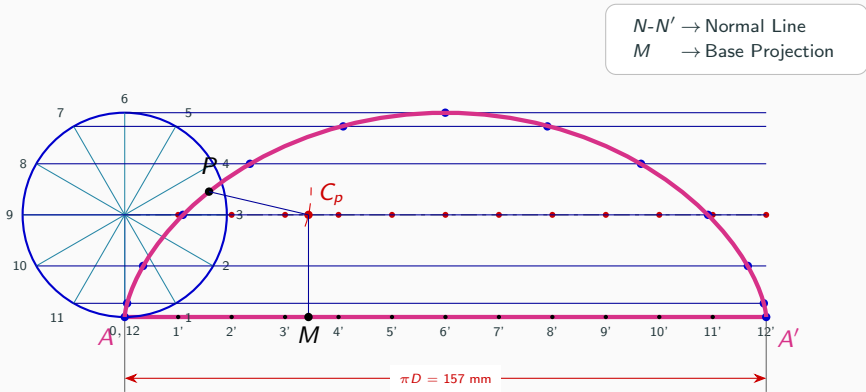
Step 8: Mark Point P & Locate Center C_p



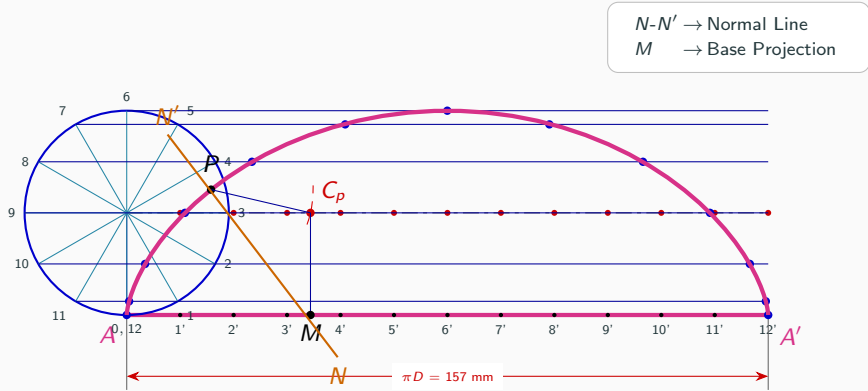
Step 8: Mark Point P & Locate Center C_p



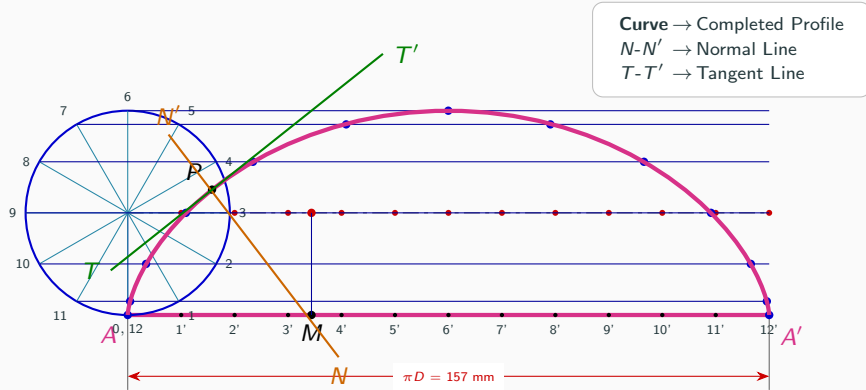
Step 9: Construct Normal Line $N-N'$



Step 9: Construct Normal Line $N-N'$



Step 10: Construct Tangent Line $T-T'$ & Final Result



Step 10: Construct Tangent Line $T-T'$ & Final Result

